

EE6203 Computer Control Systems

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LECTURE NO 1

State Space Design Methods and
Optimal Control
(15 hrs)

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References

- i. Digital Control Systems by Benjamin C Kuo, Saunders College Publishing, 1992.
- ii. Computer Controlled Systems : Theory and Design by Karl J Astrom and Bjorn Wittenmark, Prentice Hall, 1997.
- iii. Digital Control Systems Analysis and Design by Charles L Philips and H Troy Nagle, Prentice Hall, 1995.

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- iv. Digital Control of Dynamic Systems by Franklin G F, Powell J D and Workman M, Prentice Hall, 1998 (2006).
- v. Digital Control Systems : Theory, Hardware and Software by Constantine H Houpis and Gary B Lamont, McGraw Hill, 1992.

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1 State variables analysis

Objectives :

Design an algorithm for the computer such that a sequence $u(kT)$ can be generated to control the process and the output $y(t)$ vary according to some pre-specified criteria.

See Figure 1.

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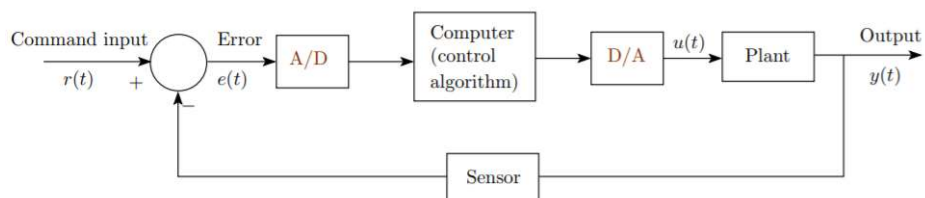


Figure 1. A Digital Control System

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Typical specifications :

- Steady-state tracking accuracy :

$$\lim_{t \rightarrow \infty} (r(t) - y(t))$$
- Transient accuracy (dynamic response) :
 - Rise time
 - Overshoot
 - Settling time
- Control effort required :
 - Maximum magnitude of $u(kT)$
 - Energy of $\sum (u^2(kT))$

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Design approaches :

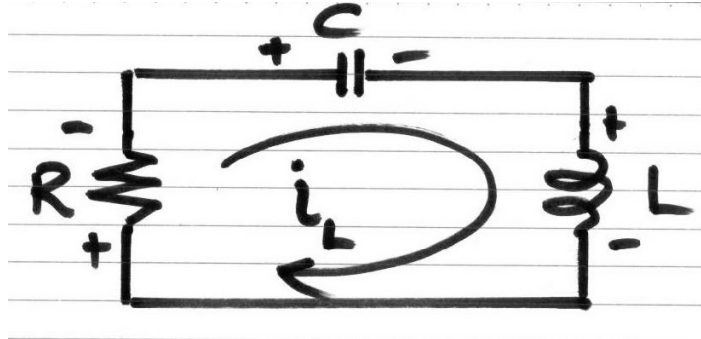
- i. Frequency domain design (classical approach)
- ii. State space design (modern approach)

Advantages of state-space approach:

- Convenient for computer applications.
- Allows a unified representation of single-variable and multi-variable systems and various types of sampling schemes.
- Can be applied to nonlinear and time-varying systems.

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Motivation : Consider the series RLC circuit as shown. By Kirchoff's Voltage Law,



$$v_L(t) + v_R(t) + v_C(t) = 0$$

$$L \frac{di_L(t)}{dt} + i_L(t)R + \frac{1}{C} \int_0^t i_L(\tau) d\tau + V_C(0) = 0$$

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$$\Rightarrow \frac{d^2 i_L(t)}{dt^2} + \frac{R}{L} \frac{di_L(t)}{dt} + \frac{1}{LC} i_L(t) = 0$$

The solution gives $i_L(t)$. What about $v_C(t)$?

- Not adequately reflect the behaviour of the circuit.
- Time consuming in finding solutions.

Two important variables : $v_C(t)$ and $i_L(t)$

Knowledge of these two variables enable the computations of other voltages in the circuit.

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Define a vector,

$$\mathbf{x}(t) = \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix} \Rightarrow \dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{v}_C(t) \\ \dot{i}_L(t) \end{bmatrix}$$

$$i_L(t) = i_C(t) = C \frac{dv_C(t)}{dt} = C \dot{v}_C(t)$$

$$\Rightarrow \dot{v}_C(t) = \frac{1}{C} i_L(t)$$

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$$\dot{i}_L(t) = \frac{1}{L} v_L(t) = \frac{1}{L} (-v_C(t) - v_R(t))$$

$$\Rightarrow \dot{i}_L(t) = -\frac{1}{L} v_C(t) - \frac{R}{L} i_L(t)$$

$$\Rightarrow \begin{bmatrix} \dot{v}_C(t) \\ \dot{i}_L(t) \end{bmatrix} = \begin{bmatrix} 0 & 1/C \\ -1/L & -R/L \end{bmatrix} \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix}$$

This is a first order matrix differential equation of the RLC circuit.

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1.1 Review of continuous-time state space representations

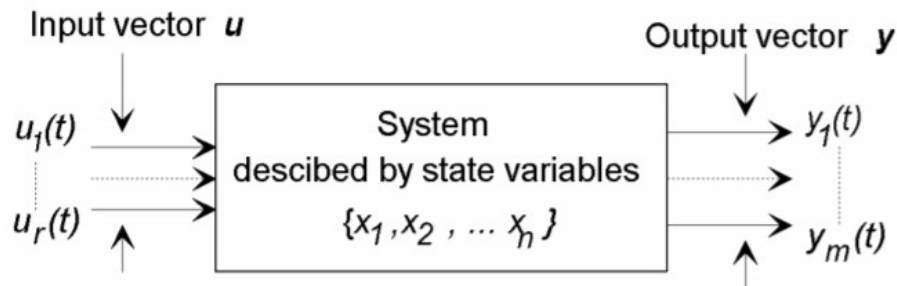


Figure 2. A LTI MIMO continuous-time system with n state variables, r inputs and m outputs

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- The state space model of a system is a mathematical description of the system in terms of a minimum set of internal variables $x_1(t), x_2(t), \dots, x_n(t)$, together with the knowledge of those variables at an initial time t_0 and the system inputs $u_i(t), j = 1, 2, \dots, r$ over $[t_0, \infty)$ are sufficient to predict the future system states and outputs for $t \geq t_0$.
- A continuous-time system can be modelled using these states by a set of first-order differential equations, called *state equations*.

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- Together with the *output equation*, a LTI continuous-time system can be represented in state space as

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad \text{.....} \quad (1.1)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad \text{.....} \quad (1.2)$$

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}; \mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_r(t) \end{bmatrix}; \mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_m(t) \end{bmatrix}$$

Example 1.1 State space description of a dynamical system

Consider the Newton's Law $M\ddot{x}(t) = F$ where M is the mass, F is force and $x(t)$ is the displacement. Define,

$$\begin{cases} x_1(t) = x(t) \\ x_2(t) = \dot{x}(t) \end{cases} \Rightarrow \begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = F/M \end{cases}$$

If the output is $y(t) = x_1(t)$, then a state space model of the system is

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1/M \end{bmatrix} F$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

1.2 Transfer function from the given state space model

Take Laplace transform of (1.1) and (1.2) with $\mathbf{x}(0) = 0$,

$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}U(s)$$

$$\Rightarrow \mathbf{X}(s) = [s\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B}U(s) \quad \dots(1.3)$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}U(s) \quad \dots(1.4)$$

$$\Rightarrow \mathbf{Y}(s) = \left(\mathbf{C}[s\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B} + \mathbf{D} \right) U(s)$$

The transfer function

$$\frac{\mathbf{Y}(s)}{\mathbf{U}(s)} = \left(\mathbf{C}[\mathbf{sI} - \mathbf{A}]^{-1} \mathbf{B} + \mathbf{D} \right) \quad \dots (1.5)$$

$$= \left(\frac{\mathbf{C} \operatorname{adj}[\mathbf{sI} - \mathbf{A}] \mathbf{B}}{\det[\mathbf{sI} - \mathbf{A}]} \right) + \mathbf{D} \quad \dots (1.6)$$

The adjoint of a matrix is the transpose of the co-factors matrix.

The denominator of the transfer function is $\det[\mathbf{sI} - \mathbf{A}]$.

Hence, the poles of the system are identical to the eigenvalues of the matrix **A**.

For single-input, single-output systems,
 $r = 1$ and $m = 1$.

Example 1.2 Obtain a state space model of the servomotor system shown in Figure 3.

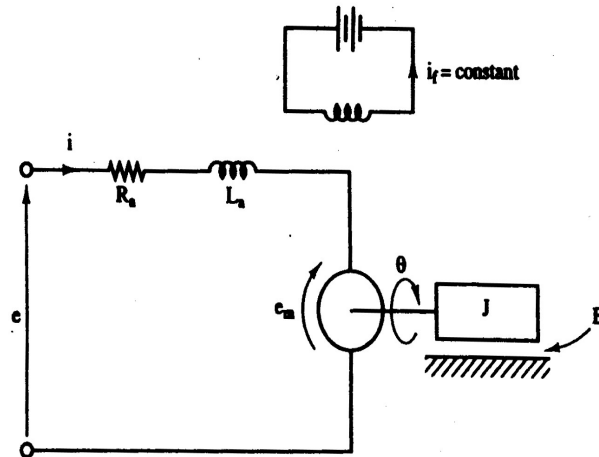


Figure 3. Servomotor system

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The motor back emf

$$e_m(t) = K_b \omega(t) = K_b \frac{d\theta(t)}{dt} \quad \dots(1.7)$$

where $\theta(t)$ - motor shaft position

$\omega(t)$ - shaft angular velocity

K_b - motor-dependent constant

If J and B are the total moment of inertia connected to the shaft and the total viscous friction, respectively, the torque developed by the motor is

$$T(t) = K_T i(t) = J \frac{d^2\theta(t)}{dt^2} + B \frac{d\theta(t)}{dt} \quad \dots(1.8)$$

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From KVL (neglect L_a)

$$e(t) = i(t)R_a + e_m(t) \quad \dots(1.9)$$

Define the following state variables and the output as the shaft position $\theta(t)$,

$$x_1(t) = \theta(t);$$

$$x_2(t) = \frac{d\theta(t)}{dt} = \dot{x}_1(t) \Rightarrow \dot{x}_2(t) = \frac{d^2\theta(t)}{dt^2}$$

$$\Rightarrow \dot{x}_2(t) = -\frac{BR_a + K_T K_b}{JR_a} x_2(t) + \frac{K_T}{JR_a} e(t)$$

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$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{BR_a + K_T K_b}{JR_a} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{K_T}{JR_a} \end{bmatrix} e(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$(1.5) \Rightarrow \frac{\Theta(s)}{E(s)} = \frac{K_T / JR_a}{s \left(s + \left(\frac{BR_a + K_T K_b}{JR_a} \right) \right)}$$

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1.3 Solution of the state vector $\mathbf{x}(t)$

The Laplace transform of (1.1) gives

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s) \quad \dots (1.13)$$

$$\Rightarrow \mathbf{X}(s) = [s\mathbf{I} - \mathbf{A}]^{-1} \mathbf{x}(0) + [s\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B}\mathbf{U}(s) \quad \dots (1.14)$$

$$\Rightarrow \mathbf{x}(t) = \mathbf{\Phi}(t)\mathbf{x}(0) + \int_0^t \mathbf{\Phi}(t-\tau)\mathbf{B}\mathbf{u}(\tau) d\tau \quad \dots (1.15)$$

$$= \mathbf{\Phi}(t)\mathbf{x}(0) + \boldsymbol{\gamma}(t), \quad t \geq 0 \quad \dots (1.16)$$

$$\boldsymbol{\gamma}(t) = \int_0^t \mathbf{\Phi}(t-\tau)\mathbf{B}\mathbf{u}(\tau) d\tau \quad \dots (1.17)$$

Alternatively,

$$\boldsymbol{\gamma}(t) = \mathcal{L}^{-1} \{ \Gamma(s) \} \quad \dots (1.18)$$

$$\Gamma(s) = [s\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B}\mathbf{U}(s) \quad \dots (1.19)$$

$\mathbf{\Phi}(t)$ is the non-singular state transition matrix and is defined as,

$$\mathbf{\Phi}(t) = \mathcal{L}^{-1} \{ [s\mathbf{I} - \mathbf{A}]^{-1} \} = e^{\mathbf{A}t} \quad \dots (1.20)$$

Some properties of $\Phi(t)$:

1. $\Phi(0) = \mathbf{I}$.
2. $\Phi(t_2 - t_1)\Phi(t_1 - t_0) = \Phi(t_2 - t_0)$.
3. $\Phi^{-1}(t) = \Phi(-t)$.
4. $\Phi^k(t) = \Phi(kT), \quad k = 0, 1, 2, 3, \dots$

Assuming $t = t_0 > 0$, substitute $t = t_0$ into (1.15) and solve for $\mathbf{x}(0)$. The resulting $\mathbf{x}(0)$ is then substituted into (1.15) to yield

$$\mathbf{x}(t) = \Phi(t - t_0)\mathbf{x}(t_0) + \int_{t_0}^t \Phi(t - \tau)\mathbf{B}\mathbf{u}(\tau) \, d\tau, t \geq t_0 \quad \dots (1.21)$$

(1.21) is known as the *state transition equation* of the system (1.1).

Example 1.3 Determine the output response of the following system to a unit step-input $u(t) = 1, t \geq 0$ and $\mathbf{x}(0) = [1 \quad -1]^T$.

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u(t)$$

$$y(t) = \mathbf{C}\mathbf{x}(t) = [1 \quad 0] \mathbf{x}(t)$$

$$[s\mathbf{I} - \mathbf{A}]^{-1} = \frac{\begin{bmatrix} s+2 & 1 \\ 0 & s \end{bmatrix}}{s(s+2)} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

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$$\Rightarrow \Phi(t) = \mathcal{L}^{-1} \left\{ [s\mathbf{I} - \mathbf{A}]^{-1} \right\} = \begin{bmatrix} 1 & 0.5(1 - e^{-2t}) \\ 0 & e^{-2t} \end{bmatrix}$$

$$(1.18) \Rightarrow \gamma(t) = \begin{bmatrix} 0.5(2t + e^{-2t} - 1) \\ 1 - e^{-2t} \end{bmatrix}$$

$$(1.16) \Rightarrow \mathbf{x}(t) = \begin{bmatrix} t + e^{-2t} \\ 1 - 2e^{-2t} \end{bmatrix}$$

$$\Rightarrow y(t) = t + e^{-2t}$$

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2 Discrete-time state space models with sample and hold

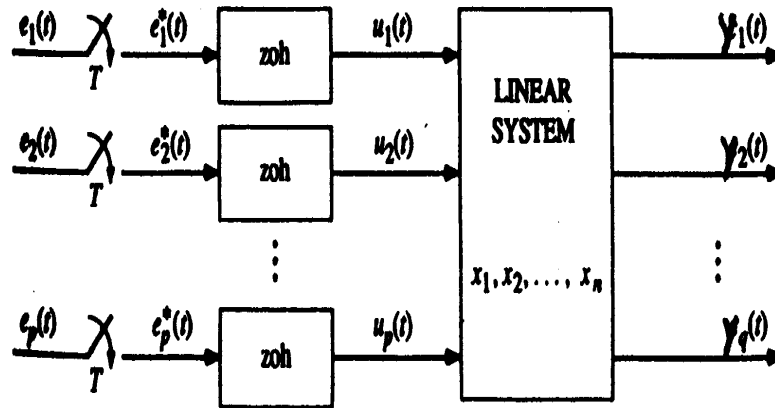


Figure 4. A LTI discrete-time system with ZOH

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The aim is to obtain the discrete state equations of the sample-data system (Figure 4) directly from the continuous-time state equations, i.e., to discretise the given continuous-time system.

The outputs of the ZOH are

$$\begin{aligned}
 u_i(t) &= u_i(kT) && \dots (2.1) \\
 &= e_i(kT), \quad kT \leq t < (k+1)T \\
 k &= 0, 1, 2, \dots, \quad i = 0, 1, 2, \dots, p.
 \end{aligned}$$

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Let the state equation and its solution be respectively, given by

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{x}(t) &= \mathbf{\Phi}(t - t_0)\mathbf{x}(t_0) \\ &\quad + \int_{t_0}^t \mathbf{\Phi}(t - \tau)\mathbf{B}\mathbf{u}(\tau) \, d\tau \quad \dots(2.2) \\ \mathbf{u}(\tau) &= \mathbf{u}(kT), \quad kT \leq \tau < (k+1)T\end{aligned}$$

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$$\begin{aligned}\Rightarrow \mathbf{x}(t) &= \mathbf{\Phi}(t - t_0)\mathbf{x}(t_0) \\ &\quad + \left(\int_{t_0}^t \mathbf{\Phi}(t - \tau)\mathbf{B} \, d\tau \right) \mathbf{u}(kT) \quad \dots (2.3) \\ \text{valid for } kT \leq t < (k+1)T. \text{ Set } t_0 &= kT. \\ \Rightarrow \mathbf{x}(t) &= \mathbf{\Phi}(t - kT)\mathbf{x}(kT) \\ &\quad + \left(\int_{kT}^t \mathbf{\Phi}(t - \tau)\mathbf{B} \, d\tau \right) \mathbf{u}(kT) \quad \dots(2.4)\end{aligned}$$

(2.4) describes $\mathbf{x}(t)$ at all times between the sampling instants kT and $(k+1)T$, $k = 0, 1, 2, \dots$ Now, let

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$$\Theta(t - kT) = \int_{kT}^t \Phi(t - \tau) \mathbf{B} d\tau \quad \dots(2.5)$$

$$\Rightarrow \mathbf{x}(t) = \Phi(t - kT) \mathbf{x}(kT) + \Theta(t - kT) \mathbf{u}(kT) \quad \dots(2.6)$$

The values of $\mathbf{x}(t)$ at successive sampling instants can be derived by setting $t = (k + 1)T$ as in the case of numerical iterations.

$$\mathbf{x}((k + 1)T) = \Phi(T) \mathbf{x}(kT) + \Theta(T) \mathbf{u}(kT) \quad \dots(2.7)$$

$$\Phi(T) = \left[\mathcal{L}^{-1} \{ [s\mathbf{I} - \mathbf{A}]^{-1} \} \right]_{t=T} \quad \dots(2.8)$$

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$$(2.5) \Rightarrow \Theta(T) = \int_{kT}^{(k+1)T} \Phi((k+1)T - \tau) \mathbf{B} d\tau$$

Set $\eta = (k + 1)T - \tau$. Then, $d\eta = -d\tau$

$$\begin{aligned} \Theta(T) &= \int_T^0 \Phi(\eta) (-d\eta) \mathbf{B} \\ &= \int_0^T \Phi(\eta) d\eta \mathbf{B} \quad \dots(2.9) \end{aligned}$$

(2.7) represents a set of first-order difference equations, referred to as the discrete state equations of the discrete-time system of Figure 4.

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Similarly, the output equation in (1.2) is discretised by setting $t = kT$,

$$\mathbf{y}(kT) = \mathbf{C}\mathbf{x}(kT) + \mathbf{D}u(kT) \quad \dots(2.10)$$

Note : Conventionally, it is common to drop "T" from the above models (2.7) and (2.10) for ease of presentations.

Example 1.4 A servomotor has a continuous-time state space representation as shown below.

Sample the system with a sampling period of $T = 0.1$ sec and obtain a discrete-time state space model of the motor.

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}(t) \end{aligned}$$

$$\Rightarrow [s\mathbf{I} - \mathbf{A}]^{-1} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+1)} \\ 0 & \frac{1}{s+1} \end{bmatrix}$$

$$\Rightarrow \Phi(t) = \mathcal{L}^{-1} \left\{ [s\mathbf{I} - \mathbf{A}]^{-1} \right\} = \begin{bmatrix} 1 & 1 - e^{-t} \\ 0 & e^{-t} \end{bmatrix}$$

$$\Rightarrow \Phi(T) = \begin{bmatrix} 1 & 1 - e^{-T} \\ 0 & e^{-T} \end{bmatrix} = \begin{bmatrix} 1 & 0.0952 \\ 0 & 0.905 \end{bmatrix}$$

$$\Theta(T) = \left[\int_0^T \Phi(\tau) d\tau \right] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.00484 \\ 0.0952 \end{bmatrix}$$

$$\Rightarrow \mathbf{x}(k+1) = \begin{bmatrix} 1 & 0.0952 \\ 0 & 0.905 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0.00484 \\ 0.0952 \end{bmatrix} u(k) \quad \dots(2.11)$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}(k) \quad \dots(2.12)$$

3 Discrete-time transfer function and state space models.

Consider,

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) \quad \dots (3.1)$$

$$\mathbf{y}(k) = \mathbf{C}\mathbf{x}(k) + \mathbf{D}\mathbf{u}(k) \quad \dots (3.2)$$

Taking z - transform and with $\mathbf{x}(0) = 0$,

$$z\mathbf{X}(z) = \mathbf{A}\mathbf{X}(z) + \mathbf{B}\mathbf{U}(z)$$

$$\Rightarrow \mathbf{X}(z) = [z\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B}\mathbf{U}(z) \quad \dots (3.3)$$

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$$\mathbf{Y}(z) = \mathbf{C}\mathbf{X}(z) + \mathbf{D}\mathbf{U}(z) \quad \dots (3.4)$$

$$= (\mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B} + \mathbf{D}) \mathbf{U}(z) \quad \dots (3.5)$$

The system transfer function is

$$\frac{\mathbf{Y}(z)}{\mathbf{U}(z)} = \mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B} + \mathbf{D} \quad \dots (3.6)$$

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3.1 Discrete time poles

A pole is a value of z such that (3.1) has a non-trivial solution when the forcing input is zero. From (3.3), this implies that

$$[z\mathbf{I} - \mathbf{A}]\mathbf{X}(z) = 0$$

has a non-trivial solution. This means that

$$\det [z\mathbf{I} - \mathbf{A}] = 0$$

i.e. the poles of the transfer function are identical to the eigenvalues of the matrix $\mathbf{A}$.

3.2 Discrete time zeros

A system zero is a value of z_0 such that the system output is zero even with a non-zero state-and-input combination.

That is, if we are able to find a non-trivial solution for $\mathbf{X}(z_0)$ and $\mathbf{U}(z_0)$ such that $\mathbf{Y}(z_0)$ is zero, then z_0 is a zero of the system.

Combining (3.3) and (3.4),

$$\begin{bmatrix} z_0 \mathbf{I} - \mathbf{A} & -\mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{X}(z_0) \\ \mathbf{U}(z_0) \end{bmatrix} = \mathbf{0}$$

which implies that the condition for the existence of non-trivial solutions is that

$$\det \begin{bmatrix} z_0 \mathbf{I} - \mathbf{A} & -\mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = 0$$

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Example 1.5 Consider

$$\mathbf{x}(k+1) = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} u(k)$$

$$y(k) = [1 \quad -1] \mathbf{x}(k)$$

$$[z\mathbf{I} - \mathbf{A}] = \begin{bmatrix} z - 1.35 & -0.55 \\ 0.45 & z - 0.35 \end{bmatrix}$$

$$\Rightarrow [z\mathbf{I} - \mathbf{A}]^{-1} = \frac{\begin{bmatrix} z - 0.35 & 0.55 \\ -0.45 & z - 1.35 \end{bmatrix}}{z^2 - 1.7z + 0.72}$$

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$$(3.6) \Rightarrow \frac{Y(z)}{U(z)} = \mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B} = \frac{1}{z^2 - 1.7z + 0.72}$$

Zeros polynomial :

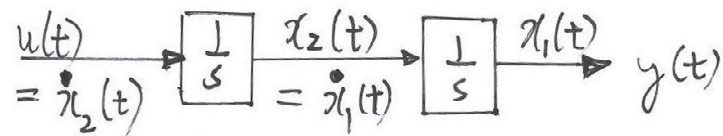
$$\left[\begin{array}{cc|cc} z\mathbf{I} - \mathbf{A} & -\mathbf{B} \\ \hline \mathbf{C} & \mathbf{D} \end{array} \right] = \left[\begin{array}{ccc|c} z-1.35 & -0.55 & -0.5 & 1 \\ 0.45 & z-0.35 & -0.5 & 0 \\ 1 & -1 & 0 & 0 \end{array} \right] = 1$$

Poles polynomial :

$$\det [z\mathbf{I} - \mathbf{A}] = z^2 - 1.7z + 0.72$$

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Example 1.6 Consider a double integrator with state variables as defined.



$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

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The discrete-time state space model :

$$[s\mathbf{I} - \mathbf{A}]^{-1} = \begin{bmatrix} s & -1 \\ 0 & s \end{bmatrix}^{-1} = \frac{1}{s^2} \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix}$$

$$\Rightarrow \Phi(T) = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}$$

$$\Theta(T) = \left[\int_0^T \Phi(\tau) d\tau \right] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}T^2 \\ T \end{bmatrix}$$

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$$\mathbf{x}(k+1) = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} \frac{1}{2}T^2 \\ T \end{bmatrix} u(k)$$

$$y(k) = [1 \quad 0] \mathbf{x}(k)$$

$$\Rightarrow \frac{Y(z)}{U(z)} = [1 \quad 0] \left[z\mathbf{I} - \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \right]^{-1} \begin{bmatrix} \frac{1}{2}T^2 \\ T \end{bmatrix}$$

$$= \frac{1}{2}T^2 \left(\frac{z+1}{(z-1)^2} \right)$$

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Example 1.7 A discrete-time system is represented by the following 4-tuple $(\mathbf{A}, \mathbf{B}, \mathbf{C}, d)$. Find the poles and zeros of the system.

$$\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{C} = [-1 \quad 1], d = 0$$

$$\det \begin{bmatrix} z\mathbf{I} - \mathbf{A} & -\mathbf{B} \\ \mathbf{C} & d \end{bmatrix} = \begin{vmatrix} z+1 & 0 & -1 \\ 0 & z+2 & -2 \\ -1 & 1 & 0 \end{vmatrix} = z$$

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$$\det [z\mathbf{I} - \mathbf{A}] = \begin{vmatrix} z+1 & 0 \\ 0 & z+2 \end{vmatrix} = (z+1)(z+2)$$

The zero is at $z = 0$ and the poles at $z = -1, -2$.

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4 The state transition equation

The most straightforward way of solving the state equation is by recursion.

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) \quad \text{.....} \quad (4.1)$$

$$k = 0,$$

$$\mathbf{x}(1) = \mathbf{A}\mathbf{x}(0) + \mathbf{B}\mathbf{u}(0) \quad \text{.....} \quad (4.2)$$

$$k = 1,$$

$$\begin{aligned} \mathbf{x}(2) &= \mathbf{A}\mathbf{x}(1) + \mathbf{B}\mathbf{u}(1) \\ &= \mathbf{A}^2\mathbf{x}(0) + \mathbf{A}\mathbf{B}\mathbf{u}(0) + \mathbf{B}\mathbf{u}(1) \end{aligned}$$

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Thus, the general solution for (4.1) given $\mathbf{x}(0)$ and $\mathbf{u}(i)$ for $i = 0, 1, \dots, (k-1)$ is

$$\mathbf{x}(k) = \mathbf{A}^k \mathbf{x}(0) + \sum_{i=0}^{k-1} \left(\mathbf{A}^{(k-i-1)} \mathbf{B} \mathbf{u}(i) \right) \quad \dots(4.3)$$

(4.3) is defined as the *state transition equation* of the discrete-time system (4.1).

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Example 1.8 Consider

$$\mathbf{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k) \dots (4.4)$$

$$y(k) = [3 \quad 1] \mathbf{x}(k) \dots (4.5)$$

$$\mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad u(k) = 1; k = 0, 1, 2, \dots$$

$$(4.3) \Rightarrow \mathbf{x}(k) = \sum_{i=0}^{k-1} (\mathbf{A}^{(k-i-1)} \mathbf{B}) \dots (4.6)$$

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$$\mathbf{x}(1) = \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$y(1) = [3 \quad 1] \mathbf{x}(1) = 1$$

$$\mathbf{x}(2) = \sum_{i=0}^1 (\mathbf{A}^{(1-i)} \mathbf{B}) = \mathbf{A} \mathbf{B} + \mathbf{B} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$y(2) = [3 \quad 1] \mathbf{x}(2) = 1$$

Hence, the states and output can be determined at successive time instants.

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